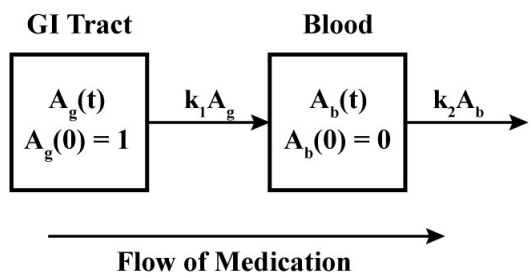
Phase 4 (part 2): First-Order Systems of ODEs

1.] In pharmaco-kinetics, "flow of medication" problems can be easily formulated to track the progression of a drug from the intestine to the bloodstream. This is called a *compartment-based model*:

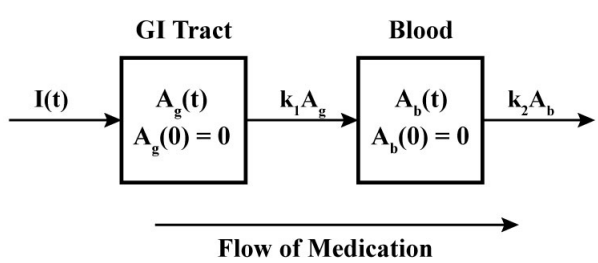
(a) Model the absorption of an antihistamine when ingested into the GI-Tract using the following diagram:



Write down the corresponding differential equation system corresponding to this model. Solve the system in Python and plot both solutions for $A_g(t)$ and $A_b(t)$ on the same figure being sure to label your figure with titles and axes appropriately. Note, k_1 and k_2 are parameters who values are up to you to decide, but to get started, define $k_1 = 0.6931$ and $k_2 = 0.0231$. What happens when you change these values?

$$\hookrightarrow \begin{cases} \frac{dA_g}{dt} = -k_1A_g & A_g(0) = 1 \\ \frac{dA_b}{dt} = k_1A_g - k_2A_b & A_b(0) = 0 \end{cases}$$

(b) Suppose there is an influx of medication every 6 hours into the GI Tract. In this scenario, there is a pulse of medication given by the function $I(t)$:



Write down the corresponding differential equation system corresponding to this model. Plot both solutions for $A_g(t)$ and $A_b(t)$ on the same figure being sure to label your figure with titles and axes appropriately. Define the function $I(t)$ using the mod function in Python to simulate a dosage every 6 hours.

$$\hookrightarrow \begin{cases} \frac{dA_g}{dt} = I(t) - k_1A_g & A_g(0) = 0 \\ \frac{dA_b}{dt} = k_1A_g - k_2A_b & A_b(0) = 0 \end{cases} \qquad I(t) = \begin{cases} 1 & \text{if } t \bmod 6 \le .5 \\ 0 & \text{if } t \bmod 6 > .5 \end{cases}$$

2.] The Lotka-Volterra equations describe a *predator-prey* relationship:

$$\frac{dx}{dt} = \alpha x - \beta xy$$
$$\frac{dy}{dt} = -\gamma y + \delta xy$$

Choose a set of parameters for this model and solve the system in Python. Plot the solutions against time. Make a second plot that plots y versus x. This is called the *Phase Plane*. Can you describe the relationship between the time series plot and the phase plot?

- Time-Series plot is periodic in nature which yields a orbit in the Phase-Plane. The two species are constantly experiencing "ups and downs" as time goes on.

3.] Below is a two-dimensional first-order system of equations that models *competing species*:

$$\frac{dx}{dt} = k_1 x \left(1 - \frac{x}{m_1}\right) - \delta_1 xy$$
$$\frac{dy}{dt} = k_2 y \left(1 - \frac{y}{m_2}\right) - \delta_2 xy$$

Choose a set of parameters for this model and solve the system in Python. Plot the solutions against time. Make a second plot that plots y versus x.

- Depending on the parameters, one out of the two species will die out while the other reaches its carrying capacity or both will coexist to an equilibrium value less than their respective carrying capacity.